

# ROWAN BRAD QUNI-GUDZINAS

Rowan Brad Quni-Gudzinis

**Research & Technology Leader — Quantum Computing · Artificial Intelligence · National-Scale Systems**

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## PROFESSIONAL SUMMARY

I design architectures, build products, and publish research at the intersection of quantum computing, artificial intelligence, and large-scale systems. Over 20 years, I've: delivered national-scale platforms (AARP Livability Index); founded an AI nonprofit featured in national media (Empowering Change); and published the quantitative case that the quantum computing industry's current trajectory is a thermodynamic dead end. I bring 310+ publications, a patent portfolio, and a rigorous methodology for epistemic hygiene audits and cross-domain innovation—most recently, 9 papers published in a single week (July 2026) spanning quantum foundations, scaffold-invariant analysis, and the adelic qubit architecture. I'm seeking a VP, Director, or Research Partner role where I can set technical vision, build teams, and turn breakthrough ideas into market reality.

## IMPACT HIGHLIGHTS

**Demonstrated quantitatively that the quantum industry's trajectory is a thermodynamic dead end—and published a viable alternative.**

Dilution refrigerators at millikelvin temperatures provide ~50 microwatts of cooling. Commercial cryocoolers at 4 Kelvin provide ~1 W—a **20,000× difference**. Published a complete roadmap for 4-Kelvin topological quantum processing using commercially available hardware and twisted high-temperature superconductors. Patents developed.

**Designed quantum hardware where fault tolerance is a property of the geometry, not a software protocol.**

Ultrametric Quantum Computing (UQC) uses a discrete hierarchical geometry where the mathematics itself prevents small errors from accumulating—in contrast to standard architectures that can require on the order of **1,000 physical qubits per logical qubit** for software-level error correction. UQC embeds suppression in the hardware, potentially eliminating this overhead. Published across 50+ documents. Patents developed.

**Led the product that scores every neighborhood in America.**

Product Manager for the AARP Livability Index: integrated **50+ data sources across 7 domains** to score every U.S. neighborhood. Full product lifecycle ownership across multiple releases. Platform and methodology cited in **20+ academic and policy studies**. Used nationwide to guide municipal planning, grant allocation, and age-friendly policy.

**Founded an AI nonprofit. Built the product. Made national news.**

Took Empowering Change from incorporation through public launch. Built an LLM-powered legal navigation platform with three integrated modules for self-represented litigants. Featured in national media for pioneering AI-driven legal democratization. Validated product design firsthand by serving as a pro se litigant in San Francisco tenant rights cases.

**Managed \$1.5M in federal R&D contracts with full procurement authority.**

Certified Contracting Officer's Representative at U.S. Department of Transportation. Full oversight of technical performance, budgets, milestones, and contractor deliverables. Led key components of a national forecasting model that informs federal infrastructure investment. Clearance-eligible.

**310+ publications, a patent portfolio, COR certification, and twenty years of sustained output across government, consulting, nonprofit, and independent research.**

Open-access body of work spanning quantum architecture, AI/ML systems, signal processing, epistemic hygiene methodology, and scaffold-invariant analysis—freely accessible on Zenodo and ResearchGate. Patent portfolio covering quantum computing architectures, neural network designs, and foundational computing paradigms.

## PROFESSIONAL EXPERIENCE

### Principal Investigator & Founder

**QWAV / QNFO**—Independent Research | 2024–Present

- **Research Output:** Published 310+ scholarly works spanning quantum computing architecture, AI/ML systems, signal processing, epistemic hygiene methodology, scaffold-invariant analysis, and cross-domain methodology—all open-access on Zenodo, ResearchGate, and SSRN. Recent output includes 9 papers in a single week (July 2026) on quantum foundations, structural formalism analysis, and the adelic qubit architecture.
- **Quantum Computing:** Published the quantitative case against dilution-refrigerated quantum architectures: ~50 microwatts of cooling at millikelvin versus ~1 W at 4 Kelvin—a  $20,000\times$  gap that fundamentally limits scalability. Published a complete 4-Kelvin topological processing roadmap using  $45^\circ$  twisted Bi-2212 superconductors (predicted thermal stability margin  $\Gamma \sim 80$ ). Designed Ultrametric Quantum Computing (UQC), embedding error suppression in the hardware geometry—potentially eliminating the 1,000:1 qubit overhead of conventional error correction. Proposed Quantum Resonance Computing (QRC), a field-based alternative paradigm. Published across 50+ documents. Patents developed.
- **Artificial Intelligence:** Proposed Prime-Attentive Neural Networks (PANNs), a neural architecture incorporating number-theoretic organizational principles. Published the Alpha Pi Project, a 7-chapter monograph adapting cardiac signal processing techniques to quantum state readout. Identified the fundamental signal-processing ceiling imposed by continuous-state architectures—a finding that directly motivated the UQC design.
- **Intellectual Property & Credentials:** Developed patent portfolio covering quantum architectures (UQC, QRC, 4K topological), neural networks (PANNs), and foundational computing paradigms.

### Executive Director & Founder

**Empowering Change**—501(c)(3) Nonprofit | San Francisco, CA | 2023–2024

- **Founded the organization and led it from incorporation through public launch.** Built an LLM-powered legal navigation platform with three integrated modules: document drafting, procedural guidance, and legal jargon translation for self-represented litigants.
- **Featured in national media** for pioneering AI-driven legal democratization.
- **Validated product design firsthand** by serving as a pro se litigant in San Francisco tenant rights cases, using direct experience to inform platform requirements.

### Product Manager & Senior Methods Advisor

**AARP Public Policy Institute** | Washington, DC | 2016–2021

- **Led the AARP Livability Index**—a national platform integrating 50+ distinct data sources across 7 domains to score every U.S. neighborhood. Owned the full product lifecycle from strategy through multiple public releases.
- **Generated measurable policy influence.** Platform and methodology cited in 20+ academic and policy studies. Used nationwide to inform municipal planning, community development grant allocation, and age-friendly policy.
- **Co-authored the Livability Index 2018 report** and multiple AARP policy briefs on housing affordability, transportation access, and age-friendly community development.
- **Managed complex multi-stakeholder relationships** spanning federal agencies, state and local governments, academic partners, advocacy organizations, and internal AARP divisions.

Senior Associate, Data Analytics

Deloitte Consulting LLP | Washington, DC | 2015–2016

- **Led data analytics and machine learning engagements** for federal and public-sector clients. Deployed predictive models and data integration strategies addressing complex organizational challenges.
- **Operated within a top-tier consulting environment**, combining technical delivery with client management, executive communication, and business development.

Data Analyst & Research Manager

U.S. Department of Transportation—Federal Highway Administration (FHWA) | Washington, DC | 2011–2015

- **Managed a federal research portfolio exceeding \$1.5M** as certified Contracting Officer’s Representative. Full authority for technical oversight, budgeting, milestone tracking, and contractor deliverable acceptance.
- **Led key technical components of the national Tour-Based Model System** for long-distance passenger travel forecasting. Model output directly informs federal infrastructure investment decisions.
- **Coordinated interagency modeling standards** across federal, state, and academic partners. Published peer-reviewed research on multimodal travel behavior and national transportation forecasting.

Earlier Professional Roles

| Role                          | Organization                                 | Period    |
|-------------------------------|----------------------------------------------|-----------|
| Platform Data Product Manager | iManage                                      | 2022–2023 |
| Senior Product Manager        | Epsilon (Publicis Groupe)                    | 2021–2022 |
| Senior Data Scientist         | The Advisory Board Company                   | 2016      |
| Consultant                    | IBI Group                                    | 2008–2011 |
| Systems Planner               | Delaware Valley Regional Planning Commission | 2005–2008 |
| GIS Analyst                   | Cross County Connection TMA                  | 2004–2005 |

TECHNICAL EXPERTISE

| Area                             | Capabilities                                                                                                                                                                                                                                  |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Quantum Computing                | Architecture design (UQC, QRC, 4K topological), cryogenic systems analysis, topological quantum computing, hardware-level error suppression, technology roadmapping, quantum–classical interfaces                                             |
| AI & Machine Learning            | Neural architecture design (PANNs), LLM deployment and productization, applied ML/NLP, predictive modeling, signal processing, AI product development                                                                                         |
| Product & Platform Leadership    | Full product lifecycle, product strategy and roadmap, multi-stakeholder management, go-to-market planning, cross-functional team building, user-centered design                                                                               |
| Data & Analytics Architecture    | Large-scale data integration (50+ heterogeneous sources), national-scale statistical modeling, geospatial analytics (GIS), real-time pipeline design, federal data standards                                                                  |
| Research & Innovation Management | Independent research program design, patent strategy, open science dissemination, technology commercialization, federal grant and contract management (COR)                                                                                   |
| Cross-Domain Methodology         | Structural pattern recognition across disparate systems, complex systems modeling, multi-scale architecture, emergence and feedback analysis, formal isomorphism identification, epistemic hygiene auditing, scaffold-invariant decomposition |

## PUBLICATIONS & INTELLECTUAL PROPERTY

- **310+ scholarly works** published open-access on Zenodo, ResearchGate, and SSRN—freely accessible, no paywalls. Works span preprints, technical reports, and books across quantum architecture, AI/ML, signal processing, epistemic hygiene methodology, and scaffold-invariant cross-domain analysis.
- **Patent portfolio** covering quantum computing architectures (UQC, QRC, 4-Kelvin topological processing), neural network designs (Prime-Attentive Neural Networks), and foundational computing paradigms.
- **Representative works:**
  - *Orchestrating the Quantum Future*—4-Kelvin quantum computing roadmap (v1.0–v1.1)
  - *Ultrametric Quantum Computation*—UQC architecture series (v0.1–v1.0, 50+ documents)
  - *Quantum Resonance Computing*—Field-based quantum paradigm proposal
  - *The Adelic Qubit*—Manufacturable hardware/software architecture specification (v1.0–v1.1)
  - *The Scaffold-Lock Hypothesis*—How notational choices marginalize formal cross-domain systems
  - *The Thermodynamic Imperative*—Quantitative analysis of quantum scaling limits
  - *The Conditional Advantage*—Epistemic hygiene audit of quantum computational advantage claims
  - *The Re-Entry Thesis*—Structural invariance of self-referential paradoxes across disciplines
  - *42 Theses on Pattern-Based Reality*—Synthesis monograph (v2.0, July 2026)
  - *Alpha Pi Project*—7-chapter monograph: cardiac signal processing → quantum architecture
  - *A General Theory of Process*—Cross-domain isomorphism framework
  - *The Information Spectrum*—Book-length treatment of information theory

## EDUCATION

- **M.S., City and Regional Studies**—Rutgers University, Edward J. Bloustein School of Planning and Public Policy (2008)
- **B.A., Urban Studies and Community Planning**—Rutgers University–Camden (2004)

## CERTIFICATIONS & PROFESSIONAL IDENTIFIERS

- **Project Management Professional (PMP)**—Project Management Institute (2016)
- **Contracting Officer’s Representative (COR)**—Defense Acquisition University (2011) · *Clearance-eligible*
- **American Institute of Certified Planners (AICP)**—American Planning Association (2009)
- **ORCID:** 0009-0002-4317-5604 · **ISNI:** 0000 0005 2645 6062

## TARGET ROLES

| Sector                                 | Representative Roles                                                                                                     | Why This Profile Fits                                                                                                        |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Quantum Computing</b>               | VP/Director of Quantum Research, Architecture, Hardware, or Engineering; Chief Quantum Scientist; Distinguished Engineer | Published architectures (UQC, QRC, 4K topological). Quantitative scaling analysis. 50+ technical documents. Patent portfolio |
| <b>AI &amp; Machine Learning</b>       | VP/Director of AI Research, Applied AI, or ML Engineering; Chief AI Scientist                                            | PANNs architecture. Launched LLM-based platform. Data science leadership at national scale. 310+ publications                |
| <b>Big Tech &amp; Platforms</b>        | Director of Research, VP Emerging Technology, Distinguished Scientist, Director of Advanced Technology                   | Cross-domain expertise (quantum + AI + systems). Product leadership at national scale. Patent portfolio                      |
| <b>Defense &amp; National Security</b> | Director of Quantum Research, Chief Scientist, VP Advanced Technology, Senior Technical Fellow or Advisor                | COR certification (clearance-eligible). Federal agency experience. Quantum architectures. Complex systems modeling           |
| <b>Government Labs &amp; FFRDCs</b>    | Director of Quantum Research, Chief Scientist, Program Director, Senior Technical Advisor                                | COR certification. Federal program management. Quantum computing R&D. Publication record                                     |
| <b>Venture Capital</b>                 | Research Partner, Technical Fellow, Entrepreneur in Residence, Director of Deep Tech Research                            | Technical evaluation (quantum + AI). 310+ publications. Founded two organizations. Product leadership                        |

| Sector                          | Representative Roles                                                                                                        | Why This Profile Fits                                                                                      |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>Consulting</b>               | Director/VP of Quantum Practice, VP of AI & Advanced Analytics, Senior Expert/Fellow, Practice Lead for Emerging Technology | Prior Deloitte experience. Client communication. 310+ publications. Cross-domain methodology               |
| <b>Startups &amp; Scale-ups</b> | CTO, Co-Founder/CEO, VP of Engineering, Head of Research, Chief Product Officer                                             | Founded two organizations. Product leadership (AARP, Empowering Change). Patent strategy. Technical vision |
| <b>Financial Services</b>       | VP of Quantum or Advanced Computing, Director of Advanced Analytics, Head of Computational Research                         | Quantum architectures for optimization and risk. Data science at scale. Systems modeling                   |
| <b>Policy &amp; Think Tanks</b> | Senior Fellow, Director of Technology or Innovation Policy, Research Director                                               | 310+ publications. Federal policy experience (AARP, FHWA). Cross-domain expertise                          |

## NOTES

- **Name History:** Formerly known professionally as Brad Gudzinias. All research since 2024 published under Rowan Brad Quni-Gudzinias.
- **Location:** Open to remote, hybrid, or on-site. Willing to relocate.